

Inversion Problem Set

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§1 Easy problems

Problem 1.1 (BAMO 2008/6). A point D lies inside triangle ABC . Let A_1, B_1, C_1 be the second intersection points of the lines AD, BD and CD with the circumcircles of BDC, CDA and ADB , respectively. Prove that

$$\frac{AD}{AA_1} + \frac{BD}{BB_1} + \frac{CD}{CC_1} = 1$$

Problem 1.2 (ISL 2003/G4). Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P , and Γ_2, Γ_4 are externally tangent at the same point P . Suppose that Γ_1 and Γ_2 ; Γ_2 and Γ_3 ; Γ_3 and Γ_4 ; Γ_4 and Γ_1 meet at A, B, C, D , respectively, and that all these points are different from P . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

Problem 1.3 (NIMO 2014). Let ABC be a triangle and let Q be a point such that $AB \perp QB$ and $AC \perp QC$. I is the incenter of ABC . It touches the sides BC, AC, AB at D, E, F respectively. If ray QI intersects EF at P , prove that $DP \perp EF$.

Problem 1.4 (Chinese MO 2006). Let $ADBE$ be a quadrilateral inscribed in a circle with diameter AB whose diagonals meet at C . Let γ be the circumcircle of $\triangle BOD$, where O is the midpoint of AB . Let F be on γ such that OF is a diameter of γ , and let ray FC meet γ again at G . Prove that A, O, G, E are concyclic.

Problem 1.5 (USMCA 2021). Three circles $\Gamma_A, \Gamma_B, \Gamma_C$ are externally tangent. The circles are centered at A, B , and C . Circles Γ_B and Γ_C meet at D , circles Γ_C and Γ_A meet at E , and circles Γ_A and Γ_B meet at F . Let GH be a common external tangent of Γ_B and Γ_C on the opposite side of BC as EF , with G on Γ_B and H on Γ_C . Lines FG and EH meet at K . Point L is on Γ_A such that $\angle DLK = 90^\circ$. Prove that

$$\frac{LG}{LH} = \frac{FG}{EH}.$$

§2 Medium Problems

Problem 2.1. Let ABC be a triangle with circumcircle γ and denote by M and N the midpoints of $\overline{AB}$ and $\overline{AC}$. Suppose the circumcircles of triangles ABN and ACM meet again at P . Prove that $\overline{AP}$ is concurrent with the tangents to γ at B and C .

Problem 2.2 (Khan Rafiul Islam Lian). Let ABC be a triangle with circumcenter O and orthocenter H . Let Γ be the circle with radius AO and center A . Γ intersect the circle BOC at F . OA intersect Γ at A' . Prove that $A'AHF$ cyclic.

Problem 2.3 (USA TST 2017/2). Let ABC be an acute scalene triangle with circumcenter O , and let T be on line BC such that $\angle TAO = 90^\circ$. The circle with diameter $\overline{AT}$ intersects the circumcircle of $\triangle BOC$ at two points A_1 and A_2 , where $OA_1 < OA_2$. Points B_1, B_2, C_1, C_2 are defined analogously.

- (a) Prove that $\overline{AA_1}, \overline{BB_1}, \overline{CC_1}$ are concurrent.
- (b) Prove that $\overline{AA_2}, \overline{BB_2}, \overline{CC_2}$ are concurrent on the Euler line of triangle ABC .

Problem 2.4. Let ABC be a triangle. I_A is its A -excenter. The bisectors of $\angle B$ and $\angle C$ intersect AC and AB at E and F respectively. EF intersects the circumcircle of (ABC) at G and H (G and C lie on the opposite sides of AB). (BEG) and (CFH) intersect at X and Y . Prove that X, Y, A, I_A .

Problem 2.5 (AoPS). Given $\triangle ABC$ with incircle ω and incenter I . $S \in AI$ such that $CS \perp AI$. $X, Y \in AB$ such that SX, SY are tangent to ω . Prove that (IXY) and (BIC) are tangent.

§3 Hard Problems

Problem 3.1 (OMMC 2023/9). Let ABC be a triangle with incircle ω . Let ω_1, ω_2 , and ω_3 be three circles centered at A, B , and C respectively tangent to ω at points D, E , and F respectively. Show there exists a circle $\Gamma \neq \omega$ tangent to circles ω_1, ω_2 , and ω_3 centered on the Euler line of $\triangle DEF$.

Problem 3.2 (ISL 2006/G6). Circles ω_1 and ω_2 with centers O_1 and O_2 are externally tangent at a point D and internally tangent to a circle ω at points E and F respectively. Line t is the common tangent of ω_1 and ω_2 at D . Let $\overline{AB}$ be the diameter of ω perpendicular to t , so that A, E, O_1 are on the same side of t . Prove that lines AO_1, BO_2, EF and t are concurrent.

Problem 3.3 (IGO 2021 Advanced/3). Consider a triangle ABC with altitudes AD, BE , and CF , and orthocenter H . Let the perpendicular line from H to EF intersects EF, AB and AC at P, T and L , respectively. Point K lies on the side BC such that $BD = KC$. Let ω be a circle that passes through H and P , that is tangent to AH . Prove that circumcircle of triangle ATL and ω are tangent, and KH passes through the tangency point.

Problem 3.4. Let ABC be a scalene acute angled triangle with orthocenter H . AT is the bisector of $\angle BAC$. P, Q lie on AC, AB respectively such that $AT \perp PQ$. (BQH) and (CPH) intersect again at X . Prove that, AT is tangent to (AHX) .

§4 Buffer Problems

Problem 4.1. Let ABC be a triangle. I_A is its A -excenter. The bisectors of $\angle B$ and $\angle C$ intersect AC and AB at E and F respectively. EF intersects the circumcircle of (ABC) at G and H (G and C lie on the opposite sides of AB). (BEG) and (CFH) intersect at X and Y . Prove that X, Y, A, I_A are concyclic.

If you want to know anything about the class or Problem set, make sure that you don't contact us.